

Advanced

Azure



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WORKFORCE DEVELOPMENT



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What is Azure?

Azure is Microsoft's cloud platform—your data center, on demand.

- **A Global Network of Microsoft-Managed Data Centers**
- **Let's you Build, Deploy, and Scale Applications, Services, and Infrastructure**
- **Pay only for what you use--compute, storage, networking, identity, and more**
- **Offers Thousands of Services: From VMs to AI to Kubernetes**
- **Accessible via Portal, CLI, APIs, or Code**

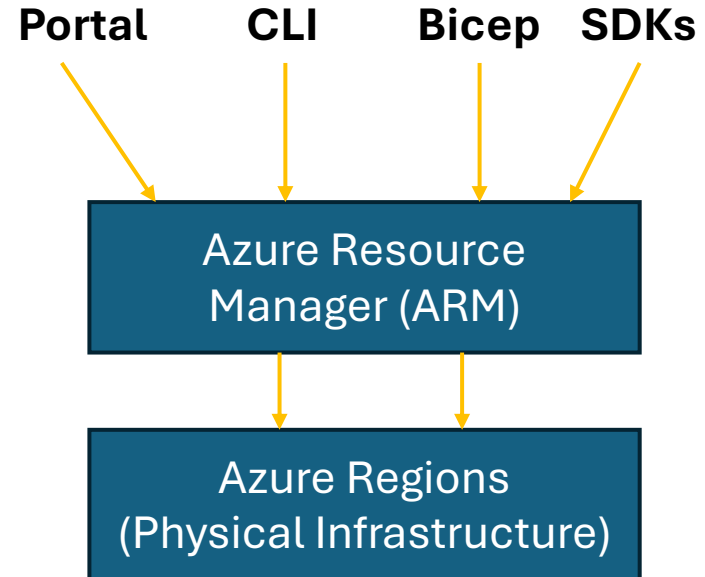
Resources On Demand

Azure is where you rent compute, storage, and services by the minute, instead of building your own data center

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The Azure Platform – A Global, Consistent Control Plane

- Azure is built on a global control plane powered by the Azure Resource Manager (ARM).
- Every service, deployment, and setting flows through ARM—regardless of the tool used.
- Regions house the physical infrastructure, but ARM provides consistent APIs and behavior everywhere.
- You can manage Azure through:
 - The Azure Portal (GUI)
 - Azure CLI and PowerShell
 - SDKs (Python, .NET, etc.)
 - Infrastructure as Code (Bicep, Terraform)

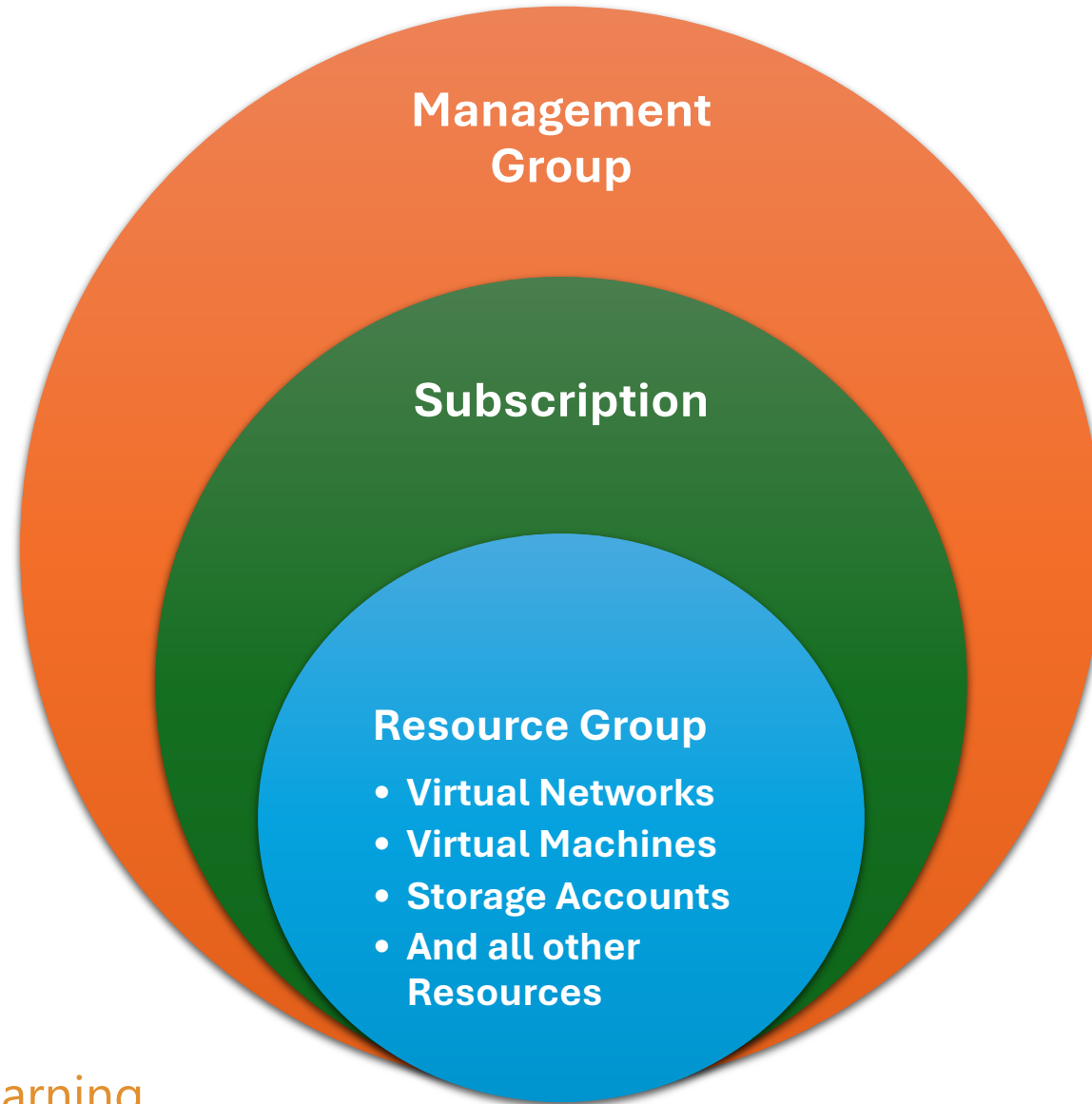


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How Azure Sees the World

- **Azure organizes your environment into hierarchical scopes:**
 - Management Groups (optional)
 - Subscriptions (Billing Boundary)
 - Resource Groups (logical containers)
 - Resources (VMs, Vnets, Storage, etc.)
- You apply RBAC, Policies, and Tags at any Scope Level
- ARM Provides declarative deployment and ensures resources are deployed in the correct order.
- Tools like Bicep let you describe your desired state and let Azure handle the rest.

Resources, Groups, and Scopes



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Azure Resource Manager (ARM)

Brand new JSON-based API that is built on resource providers

Resources belong to a resource group and share a common life-cycle

Cloud services not used for IaaS (but still for PaaS)

JSON-based deployments are declarative, idempotent and understand dependencies between resources to govern creation order

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Azure Resource Manager (ARM)

Massive
parallelism

Tags used
to organize
resources which
are also visible
in billing

Can also use
PowerShell and
the modern
Azure Portal

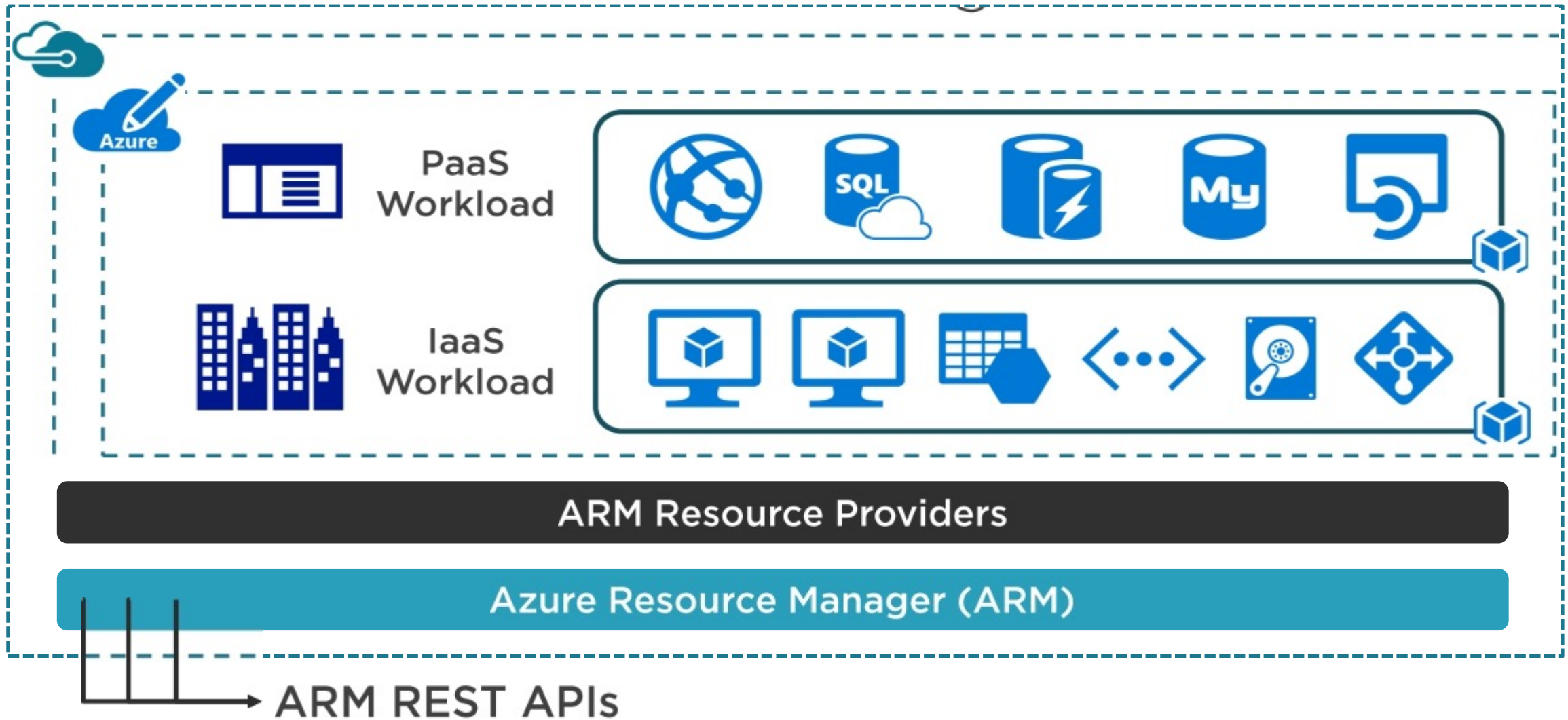
Resource Groups

- Every ARM resource exists in one, and only one, resource group
- While resource groups are created in a region they can contain resources from outside the region
- Resources can be moved between resource groups

Resource Groups

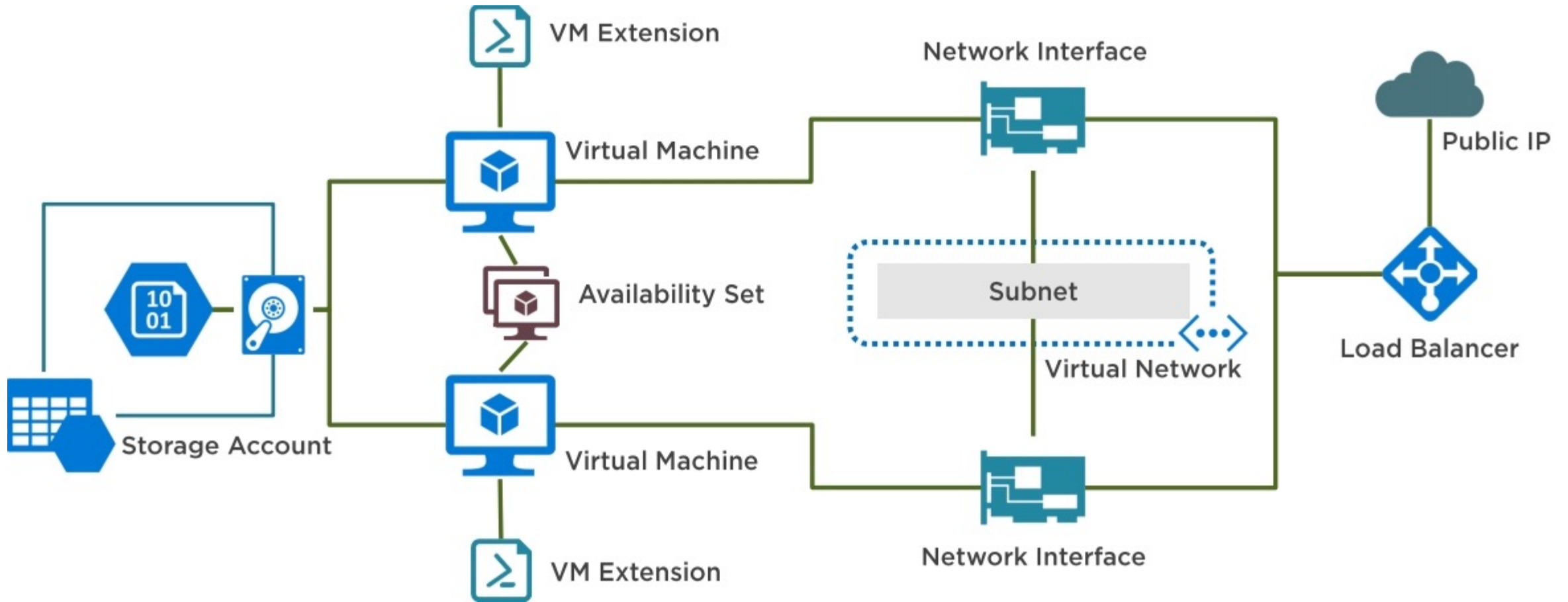
- All resources in your group should share same lifecycle.
 - Deploy, Update, and Delete together
- Resource groups can be heterogeneous or homogeneous
- Resource groups are not a boundary of access

Azure resource manager architecture



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Resources and Dependencies



Resource Group Limits

- 800 resource groups per subscription
- 800 resources per resource type per resource group
- 15 tags per resource group
- Bound by subscription

LAB: Deploy a VM via the Portal Interface

Get Started with Azure Portal!

- Traditionally, we'd build resources using Bicep Templates, or IaC tools like Terraform
- To cement our understanding of the tool, we'll use the Portal to provision a VM.
- To build a VM, we'll also need to deploy its dependencies:
 - **Resource Group**
 - **Virtual Network**
 - **Subnet**
 - **Network Interface (NIC)**
 - **Storage / Disks**
 - **Virtual Machine Image**

Pro Tip for IaC:

When writing a Bicep or ARM Template, you must declare and manage these dependencies in the correct order--often using *dependsOn* explicitly or letting Azure infer it through resource references.

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